



# APPENDIX B

## Code Metadata



First learn computer science and all the theory. Next develop a programming style.

Then forget all that and just hack.

— *George Carrette* —



## B.1 | Code Versions

| Code Name  | Version | Reference                              |
|------------|---------|----------------------------------------|
| HOKI       | 1.7.1   | <a href="#">Stevance et al. (2020)</a> |
| KAITIAKI   | 1.0.0   | This work                              |
| MATPLOTLIB | 3.8.2   | <a href="#">Hunter (2007)</a>          |
| NUMBA      | 0.58.1  | <a href="#">Lam et al. (2015)</a>      |
| NUMPY      | 1.26.3  | <a href="#">Harris et al. (2020)</a>   |
| PANDAS     | 2.1.4   | <a href="#">McKinney (2010)</a>        |
| PYTHON     | 3.11.5  | -                                      |
| SCIPY      | 1.11.4  | <a href="#">Virtanen et al. (2020)</a> |
| TAKAHE     | 1.0.0   | <a href="#">Richards et al. (2023)</a> |
| TUI        | 8       | <a href="#">Ghodla et al. (2023)*</a>  |

**Table B.1:** The code versions we use in this thesis. \*TUI was described in earlier papers, but [Ghodla et al. \(2023\)](#) was the first to name it.

## B.2 | Parameter Configuration

In this appendix we list the default parameters chosen for our binary evolution runs, as detailed in subsection 2.6.1.

The possible mass loss prescriptions (IML1/IML2) are:

0. No mass loss
1. [Reimers \(1975\)](#)
2. [Blöcker \(1995\)](#)
3. [Vassiliadis & Wood \(1993\)](#)
4. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with interpolation errors described in [Richards et al. \(2024\)](#)
5. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) (described in [Eldridge et al., 2017](#))
6. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with slowly turning on WR rates described in [Richards et al. \(2024\)](#)
7. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with no WR rates, described in [Richards et al. \(2024\)](#)
8. Unimplemented

| Parameter | Value     | Comment        | Parameter | Value     | Comment | Parameter | Value     | Comment        |
|-----------|-----------|----------------|-----------|-----------|---------|-----------|-----------|----------------|
| NM2       | 199       | Meshpoints     | ICL       | 1         |         | IT1       | 40        |                |
| ION       | 2         |                | IT2       | 20        |         | IAM       | 1         |                |
| JIN       | 15        |                | IOP       | 5         |         | OUT       | 15        |                |
| INUC      | 0         |                | NCH       | 3         |         | IBC       | 0         |                |
| JP        | 1         |                | ICN       | 0         |         | ITH       | 1         |                |
| IML1      | 5         | Mass loss (*1) | IX        | 1         |         | IML2      | 5         | Mass loss (*2) |
| IY        | 1         |                | ISGTH     | 0         |         | IZ        | 1         |                |
| IMO       | 0         |                | IMODE     | 2         |         | IDIFF     | 0         |                |
| NWRT1     | 100       |                | EPS       | $10^{-6}$ |         | NWRT2     | 1         |                |
| DEL       | 0.01      |                | NWRT3     | 2         |         | DHO       | $10^{-8}$ |                |
| NWRT4     | 1         |                | DT3       | 0         |         | NWRT5     | 0         |                |
| DDD       | 1.0       |                | NSAVE     | 10        |         | NMONT     | 0         |                |
| DT1       | 0.1       |                | DT2       | 1.2       |         | CT1       | 3.0       |                |
| CT2       | 3.0       |                | CT3       | 0.05      |         | CT4       | 0.01      |                |
| CT5       | 0.6       |                | CT6       | 0.35      |         | CT7       | 0.45      |                |
| CT8       | $10^{-4}$ |                | CT9       | 0.1       |         | CT10      | 0.3       |                |

**Table B.2:** Our general initial conditions. Unless otherwise described, these are assumed to be enabled for all runs.

| Parameter | Value            | Comment | Parameter | Value           | Comment | Parameter | Value  | Comment          |
|-----------|------------------|---------|-----------|-----------------|---------|-----------|--------|------------------|
| ZS        | 0.02             |         | ALPHA     | 1.925           |         | CH        | 0.7    |                  |
| CC        | 0.173            |         | CN        | 0.053           |         | CO        | 0.482  |                  |
| CNE       | 0.099            |         | CMG       | 0.038           |         | CSI       | 0.08   |                  |
| CFE       | 0.072            |         | RCD       | 10 <sup>6</sup> |         | OS        | 0.12   |                  |
| RML       | 0                |         | RMG       | 0               |         | ECA       | 0      |                  |
| XF        | 0.1              |         | DR        | 0.001           |         | RMT       | 1000.0 |                  |
| RHL       | 0                |         | AC        | 0               |         | AK1       | 1.0    |                  |
| AK2       | 10 <sup>-4</sup> |         | ECT       | 0               |         | TRB       | 0      |                  |
| IRAM      | 1                |         | IRS1      | 1               |         | VROT1     | 0      |                  |
| IRS2      | 1                |         | VROT2     | 10.0            |         | FMAC      | 1.0    |                  |
| FAM       | 0                |         | IVMC      | 0               |         | TRC1      | 700.0  |                  |
| IVMS      | 0                |         | TRC2      | 0               |         | MWTS      | 0      |                  |
| IAGB      | 0                |         | ISGFAC    | 0               |         | FACSGMIN  | 0.01   |                  |
| SGTHFAC   | 0                |         | ISTART    | 0               |         | HKH       | 1.0    |                  |
| GFF       | 10.0             | Removed | NNMOD     | 0               |         | CEPR      | 0      | CEE prescription |
| ALPHACE   | 1.0              |         | -         | -               | -       | -         | -      | -                |

**Table B.2:** Continued

9. ZAMS evolution to a specific ZAMS mass, custom routine by Jan Eldridge